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# Advances in forecasting realized volatility: a review of methodologies

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policymaking. Traditionally, academic research has emphasized linear models for forecasting realized volatility. However, advances in computational power have enabled the development of more sophisticated approaches, including machine and deep learning models. Despite these advancements, no comprehensive survey currently compares traditional linear methods with newer models in the context of realized volatility forecasting. This survey addresses that gap by analyzing all models used to forecast realized volatility in academic literature from 2000 to the first half of 2024. It highlights key academic contributions and examines the empirical characteristics of realized volatility. Our findings indicate that the top-performing hybrid convolutional neural network and long short-term memory model surpasses other models in forecasting accuracy. These results highlight the ability of such models to capture a broader range of factors driving realized volatility, affirming their valuable role in quantitative modeling.


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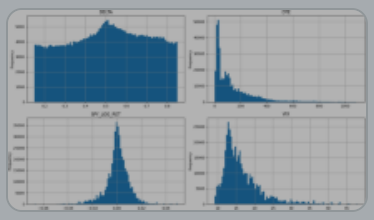

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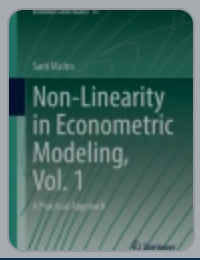


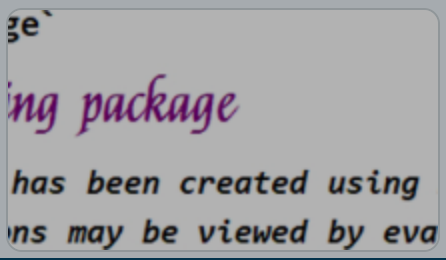


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## Introduction to stock market volatility

Over the past century, advances in digitalization, information processing, and financial modeling have transformed the financial industry from a network of small local banks into a complex global system with significant influence on the world economy (Liermann and Stegmann, 2019, p. 1). At the core of this system are the stock and derivatives markets, with the latter having an estimated notional value nearly 10 times greater than the global gross domestic product (Maverick 2024).

Given the massive volume of capital traded in the derivatives market, it is critical for financial institutions (e.g., hedge funds, trading firms, insurers, and banks) to accurately price the derivatives they trade. A key variable in pricing these instruments is the volatility of the underlying asset. Volatility reflects the distribution of an asset's returns and captures the degree of price fluctuation over time. Higher volatility signals greater uncertainty and risk, as prices are more likely to experience significant swings. Consequently, volatility plays a


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reviews and compares all models used in the academic literature to forecast realized volatility from 2000 through the first half of 2024. It is the first comprehensive review to evaluate traditional linear models alongside more advanced approaches, including convolutional neural networks (CNNs), long short-term memory (LSTM) models, heterogeneous autoregressive neural networks (HARNets), and transformers. The survey examines the design and empirical performance of each model using standard goodness-of-fit metrics. It also identifies research gaps and recommends directions for enhancing volatility forecasting techniques. Throughout the analysis, we address two central questions:

1. Which forecasting models are most valuable to academic researchers forecasting realized volatility, and how do they differ in their assumptions and designs?
2. What key empirical characteristics of realized volatility do these models address and how effectively do they capture these features?

The remainder of Section "Introduction to Stock Market Volatility" discusses the theoretical foundations of (realized) volatility, outlines its empirical properties, and describes the methodology of this survey and related work. Section "Analysis of applied forecasting models" reviews the academic literature on volatility forecasting models. Section "Conclusion" summarizes the findings, identifies gaps in current knowledge, and suggests future research directions to improve forecasting performance.


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exponential value, meaning that at time *t*, $S_t$, follows a log-normal diffusion process represented by

$$\begin{aligned} \dfrac{dS_t}{S_t} = \mu dt + \sigma dW_t. \end{aligned}$$

(1)

Here, $dW_t$ denotes the Brownian motion process, which captures the inherent randomness of asset movement. $\mu$ represents the expected instantaneous return, adjusted for dividend yield, and $\sigma$ denotes the instantaneous return volatility. Using Ito's lemma, the change in the logarithm[1] of the asset price, $S_t$, is given by

$$\begin{aligned} d \log (S_t) = \left( \mu - \frac{1}{2} \sigma ^2 \right) dt + \sigma dW_t. \end{aligned}$$

(2)

The expected return, $\mu$, may also be time-dependent, denoted $\mu _t$, without loss of generality. $\sigma$ again represents volatility, and $\sigma ^2$ denotes variance. A key assumption in this model is that volatility $\sigma$ remains constant over time.


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$$\begin{aligned} d \sigma ^2_t = \alpha (\sigma ^2_t) dt + \beta (\sigma ^2_t) dZ_t. \end{aligned}$$

(3)

Here, $dZ_t$ is a Brownian motion independent of the previously defined $dW_t$. This formulation allows Eqs. (1) and (2) to reflect time-varying volatility $\sigma _t$ instead of the constant $\sigma$. Functions $\alpha (\sigma ^2_t)$ and $\beta (\sigma ^2_t)$ are typically assumed to satisfy the following conditions:

1. Both exhibit locally bounded path variation over interval $[0, \infty )$.
2. Both satisfy Lipschitz continuity.

These relatively mild assumptions still permit volatility $\sigma _t$ to exhibit jumps, long memory, and non-stationarity without requiring an unconditional mean (Barndorff-Nielsen and Shephard 2002). Notably, this flexible formulation does not demand detailed prior knowledge of the stochastic processes governing $\alpha (\sigma ^2_t)$ and $\beta (\sigma ^2_t)$. Instead, they can be treated as bounded and Riemann integrable. Under these assumptions, $\sigma ^2_t$ is considered a semimartingale with a continuous local martingale component (Barndorff-Nielsen and Shephard 2002). The presented SV


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## Realized volatility

Although theoretically important, the integrated variance $\sigma ^2_{0,T}$ cannot be directly observed in practice. This limitation arises because it is a continuous-time measure that would require information about the asset's instantaneous volatility at every moment. In reality, only discrete and finite market data points are available. The resolution of real-world financial data is constrained by both trading hours and transaction frequency, making it impossible to capture every instantaneous volatility fluctuation.

This lack of continuous data renders the exact calculation of integrated variance, and thus volatility, impractical. Therefore, accurate approximations are necessary. The most widely used approximation is the realized variance, which is computed from high-frequency return data. Unlike theoretical measures, realized variance is observable and captures actual market behavior, making it more suitable for empirical financial modeling. It serves as a bridge between continuous-time theoretical constructs and discrete-time real-world observations. Mathematically, the realized variance $\tilde{\sigma }^2_t$ at time $t$ over a fixed interval divided into *N* subintervals is given by

$$\begin{aligned} \tilde{\sigma }^2_t = \sum _{n=1}^{N} r_{t+\frac{n}{N}}^2, \end{aligned}$$


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variance, $\sigma^2_{0,T}$. Because it is observable, realized variance enables the application of quantitative models without treating volatility as a latent variable.

Importantly, realized variance $\tilde{\sigma}^2_t$ is a robust and consistent estimator of integrated variance, as it asymptotically converges to $\sigma^2_{0,T}$. This convergence is demonstrated in Barndorff-Nielsen and Shephard (2002). As the expected value of realized variance approaches the true integrated variance, realized volatility likewise converges to true volatility. With this theoretical groundwork in place, we now turn to the empirical characteristics of realized volatility, which offer valuable insight into actual market behavior.

## Properties of volatility

This section outlines the empirical characteristics of realized volatility as identified in academic research. These characteristics help clarify the assumptions underlying the models discussed in the next section and serve as benchmarks for developing new models that more accurately reflect market behavior.

Financial econometrics, risk management, and asset pricing literature highlight six key empirical characteristics of volatility:


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closely tied to price *jumps*, where sudden news or shocks cause large movements that amplify these fat tails.

C4 Mean reversion (baseline level): Despite short-term clustering and autocorrelation (C1), volatility exhibits mean reversion over longer periods. This means it tends to fluctuate around a long-term average or baseline level (Bali and Demirtas 2008).

C5 Seasonality (periodic cycles): Volatility displays predictable seasonal patterns driven by cyclical factors such as earnings announcements, holidays, or fiscal reporting dates. As a result, some assets experience regular shifts in volatility during specific times of the year (Richter and Sørensen 2002).

C6 Co-movement (cross-asset correlation): Asset returns—and hence their volatilities—often move in tandem with other assets (Baker and Wurgler 2012). This co-movement arises during shared macroeconomic shocks or systemic market events and highlights the interconnectedness of financial instruments.

With these characteristics in mind, the following subsection explores how researchers have applied them in forecasting models. We review key modeling approaches and their theoretical underpinnings drawn from the academic literature.


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Q1. Which forecasting models are most valuable to academic researchers forecasting realized volatility, and how do they differ in terms of their assumptions and design?

Q2. What key empirical characteristics of realized volatility do these models address and how effectively do they capture these features?

Several prior studies (e.g., Kambouroudis et al. 2021; Wang et al. 2016; Wong et al. 2016; Bucci 2017) have compared different forecasting models. However, most focused solely on linear models. Other studies introduced a single new ML or DL model and compared it only with a basic linear benchmark, without offering a broad cross-class evaluation. This survey addresses that gap by providing a systematic comparison across all model classes covered in academic literature, including both traditional linear approaches and advanced ML and DL architectures such as CNN-LSTM, HARNets, and transformers. We assess each model's strengths and limitations and analyze how well they capture the key characteristics of realized volatility. This review is intended to present a complete picture of how various models perform across different empirical settings.

## Analysis of applied forecasting models

Next, we examine 32 linear, ML, and DL models that have been employed in publicly


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## Linear models

The first class of models (linear) is the oldest and most widely used in academic literature for forecasting realized volatility. Within this class, the two primary types are autoregressive moving average (ARMA) and heterogeneous autoregressive (HAR) models.

### ARMA models

ARMA employs past time series values (autoregressive) to account for autocorrelations (C1) and past forecast errors (moving average) to account for mean reversions (C4). Thus, it models weakly stationary time series using two polynomials. The standard ARMA($p$,$q$) formulation for realized volatility $\tilde{\sigma }_t$ is

$$\begin{aligned} \tilde{\sigma }_t = c + \sum _{i=1}^{p} \alpha _i \tilde{\sigma }_{t-i} + \sum _{j=1}^{q} \beta _j \epsilon _{t-j} + \epsilon _t, \end{aligned}$$

(7)

where $p$ and $q$ denote the autoregressive and moving average orders, respectively, $\epsilon _t$ is the error term at times $t$ and $c$, $\alpha _i$ for $i=1,\ldots ,p$, and $\beta _j$ for $j=1,\ldots ,q$ are the estimated coefficients. The error term is assumed to be normally distributed with mean zero and variance $E$ (i.e., $\varepsilon _t \sim N(0,E)$).


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A key assumption of the ARMA model is the weak stationarity of $\tilde{\sigma }_t$, which does not hold for all assets (Mills and Mills 2015). Weak stationarity requires that the series have constant mean and variance over time and that covariances depend only on the time lag, not the specific time. However, many financial series are nonstationary (Cavaliere and Taylor 2007), violating this assumption. To address this, the autoregressive integrated moving average (ARIMA) model introduces differencing to model the $d^{th}$ order change in volatility instead of volatility itself. This stabilizes the mean by removing trends. The differencing operator, $\Delta ^d$, is defined as

$$\begin{aligned} \Delta ^d \tilde{\sigma }= {\left\{ \begin{array}{ll} \tilde{\sigma }_t - \tilde{\sigma }_{t-1} & \text {if } d = 1 \\ \Delta (\Delta ^{d-1} \tilde{\sigma }_t) & \text {if } d> 1 \end{array}\right. } \end{aligned}$$

(8)

where $d \in \mathbb {Z}$. The ARIMA($p$,$d$,$q$) model can then be expressed as

$$\begin{aligned} \Delta ^d \tilde{\sigma }_t = c + \sum _{i=1}^{p} \alpha _i \Delta ^d \tilde{\sigma }_{t-i} + \sum _{j=1}^{q} \beta _j \epsilon _{t-j} + \epsilon _t. \end{aligned}$$


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models such as autoregressive fractionally integrated moving average (ARFIMA) and GARCH for specific indices, including the NASDAQ100.

One limitation of the ARIMA model is its inability to capture long-term dependencies, which are prevalent in volatility series (C4). This limitation is addressed by the ARFIMA model of Granger and Joyeux (1980), which uses fractional differencing to account for long memory. The model retains the ARIMA (9) structure but redefines $\Delta ^d$ as

$$\begin{aligned} \Delta ^d \tilde{\sigma }_t = \sum _{k=0}^{\infty } (-1)^k \left( {\begin{array}{c}d\\ k\end{array}}\right) \tilde{\sigma }_{t-k}. \end{aligned}$$

(10)

Here, $d \in \mathbb {N}$ is fractional. When *d* is an integer ($d \in \mathbb {Z}$), the ARFIMA model reduces to ARIMA. The infinite sum captures long-term dependencies, making ARFIMA especially effective for modeling mean-reverting behavior (C4).[4]

Empirical results from Pong et al. (2004) indicate that ARFIMA models outperform implied volatility models in $R^2$, particularly for pound and yen volatility forecasts at 1-day and 1-week horizons. Ishida and Watanabe (2009) also demonstrated strong performance from


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(11)

Empirical studies confirm the benefits of this transformation. Alizadeh et al. (2002) showed improved RMSE performance in forecasting U.S. dollar exchange rate volatility, and Koopman et al. (2005) reported better MSE and $R^2$ outcomes for the S&P 500 between October 2001 and November 2003.

Another extension involves including asymmetric realized volatility terms that vary by asset return direction. This asymmetric ARFIMA approach captures differences in volatility during declines versus rallies. Degiannakis (2008) applied such models to CAC40 and DAX30 intraday volatility, successfully capturing several empirical features. Likewise, Kambouroudis et al. (2016) found these models minimized forecast error (i.e., MAE, RMSE, adjusted $R^2$) for the S&P 500 and DJIA.

Implied volatility can also serve as a regressor. Kambouroudis et al. (2016) found that while implied volatility alone may underperform, its inclusion improves model accuracy. Bazhenov and Fantazzini (2019) corroborated this, reporting significant MSE improvements across stocks and models. Similarly, in a multivariate setting, Chen et al. (2010) found that including implied volatility in vector-ARFIMA (VARFIMA) models improved forecast accuracy as measured by MSE and MAE.


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$(i=1,\ldots ,p)$ and $(B_j)$ for $(j=1,\ldots ,q)$ are of dimension $(N\times N)$.

Empirical results from Chen et al. (2010) indicate that VARFIMA models outperform GARCH(1,1), its extensions such as FIGARCH(1,1), and basic SV models. This underscores the importance of accounting for co-movement in realized volatility forecasting. Furthermore, Andersen et al. (2003) find that the VARFIMA class of models yields significantly more accurate volatility forecasts, as measured by $(R^2)$, compared to conventional GARCH and related models.

### HAR model

The second type of linear model, HAR, introduced by Corsi (2009), represents a major advancement in volatility modeling. As noted, financial time series often exhibit features such as long memory (C4) and leptokurtosis (C3) (Cheng and Hung 2011). These features are challenging for traditional models like ARMA or vector-ARMA to capture effectively. Drawing on the heterogeneous market hypothesis of Müller et al. (1997), Corsi (2009) argues that volatility emerges from the actions of different market participants operating over varied time horizons. His model attempts to incorporate these dynamics to better represent long-memory behavior in realized volatility.

Unlike ARMA or VAR models, the HAR model decomposes realized volatility into additive


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$$\begin{aligned} \tilde{\sigma }_t = \beta _0 + \beta _{D}\tilde{\sigma }_{t-1}^{(D)} + \beta _{W}\tilde{\sigma }_{t-1}^{(W)} + \beta _{M}\tilde{\sigma }_{t-1}^{(M)} + \varepsilon _t, \end{aligned}$$

(13)

where $\tilde{\sigma }_{t-1}^{(X)} = \sqrt{\frac{1}{N_X}\sum ^{N_X}_{i=1} \tilde{\sigma }^2_{t-i}}$ for $X \in [D, W, M]$, with $N_X \in [1,5,22]$, and $\varepsilon _t \sim N(0,E)$.

Like ARFIMA models, the HAR model's linearity allows for analytical tractability and straightforward interpretation. However, it is considerably more efficient: it captures persistence with only three terms, whereas an ARMA model would need up to 22 lags to do so. This efficiency marks a significant step forward in forecasting realized volatility.

Empirical evidence suggests that the HAR model outperforms traditional models such as ARFIMA and exponential GARCH (EGARCH) in forecast accuracy. For example, Xiao et al. (2024) compared ARFIMA, HAR, and EGARCH models in predicting idiosyncratic volatility (IVOL), defined as the realized volatility not explained by market-wide movements. Their findings show that the HAR model delivers superior IVOL forecasts. Additionally, Xiao et al.


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Izzeldin et al. (2019) reported similar results when comparing HAR and ARFIMA models for U.S. stock volatility across various sectors from 2000 to 2010. They noted that ARFIMA's memory parameter, \(\Delta ^d\), was sensitive to market conditions and sampling frequency, which reduced its predictive accuracy. In contrast, HAR model forecasts remained stable across forecasting horizons, with minimal shifts in sector rankings.

The results from these studies underscore the robustness of the HAR model, which consistently delivers superior forecast accuracy across different markets and time periods. This performance likely stems from its ability to effectively capture persistent dependencies in realized volatility, something traditional models often struggle with Degiannakis et al. (2022). Nonetheless, academic literature has introduced several extensions to the HAR model that further enhance its predictive capabilities.

Many of these extensions mirror those used in ARFIMA models, demonstrating their effectiveness. First, modeling the logarithm of volatility rather than volatility itself helps address leptokurtosis (C3) and produces more accurate and theoretically sound forecasts. Second, transforming the HAR model into a multivariate framework enables it to capture dependencies and spillovers between assets, improving its ability to represent co-movement (C6) and enhancing predictive performance (Souček and Todorova 2013). This is supported by Čech and Barunik (2017), who introduced the multivariate GHAR model. Their findings


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An additional extension to the HAR model is proposed by Andersen et al. (2007), who introduce a jump regressor component. This version, referred to as the jump-HAR (J-HAR) model in this survey, includes a jump regressor defined as the excess realized variance over the realized bi-power variation.[8] This component helps to capture the leptokurtosis (fat tails) characteristic (C3) of realized volatility. Their findings show that incorporating this jump component significantly improves forecasting accuracy, particularly for long-term monthly forecasts. For the S&P 500 index and T-Bonds, the \(R^2\) values for the J-HAR model were three times higher than those of the traditional HAR model.

Another extension involves using separate HAR regressors for negative and positive returns. This version, referred to as the asymmetric-HAR (A-HAR) model in this survey, addresses the asymmetry property (C2) of realized volatility by accounting for different volatility behavior during market upturns and downturns (Dzieliński et al. 2018). To address this asymmetry, Patton and Sheppard (2015) propose a set of models based on signed realized volatilities. Their results indicate that past negative return volatility has a stronger influence on future volatility than positive return volatility. Moreover, they find that the impact of volatility jumps on future volatility depends on the sign of the jump; negative jumps lead to substantially higher future volatility than positive ones. A simple jump-A-HAR (JA-HAR) model incorporating these findings achieves significantly higher out-of-sample forecast accuracy than traditional HAR models, with forecasting horizons ranging from 1 day to 3


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improving forecast accuracy. The resulting IV-HAR model consistently outperforms all other HAR variants without IV across time and markets. However, they also found that no single IV-HAR extension dominates in all markets. For instance, including overnight returns as a regressor enhances performance in most markets, while the inclusion of realized volatility of volatility is only beneficial for two US indices. The inclusion of asymmetry (leverage) effects yields mixed results, improving forecast accuracy in only five markets.

The next subsection introduces ML models, which build on concepts from the linear modeling domain but apply more advanced techniques.

## Machine learning models

This section introduces ML techniques, which often extend the ideas underlying linear models (e.g., HAR) but employ more sophisticated quantitative methods. First, we discuss regularized regressions as a direct extension of the HAR framework. Second, we introduce kernel regression, the first non-parametric method analyzed in this survey. Finally, we explore tree-based models.

### Regularized regression

A natural extension of the HAR model is the incorporation of regularization terms. Regularized regression models, while still linear, include many dependent variables and


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$$\begin{aligned}&\text {Ridge}: ~~~~~~~ \quad C = \lambda \sum ^p_{i=1} \beta ^2_i\end{aligned}$$

(15)

$$\begin{aligned}&\text {Elastic Net}: \quad C = \sum ^p_{i=1} \lambda _1 |\beta _i| + \lambda _2 \beta ^2_i \end{aligned}$$

(16)

where \(\lambda\), \(\lambda _1\), and \(\lambda _2\) are hyperparameters typically optimized via cross-validation, and \(\beta _i\) for \(i=1,\ldots ,p\) are the linear model parameters. All three methods aim to regularize the regression model, with elastic net serving as a hybrid of LASSO and ridge regression.

In the context of realized volatility forecasting, these regularized models generalize the HAR and ARFIMA frameworks, both of which approximate an \(AR(\infty )\) process. Consequently, they are capable of capturing the same fundamental properties of realized volatility as the HAR model (i.e., autocorrelation (C1) and mean reversion (C4)). However, \(AR(\infty )\) models are typically approximated by finite-order models such as $AR(k)$ for large $k$ (Audrino and Knaus 2016). Under this approach, regularized regression models retain


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structure (1, 5, 22). This result implies that the true volatility dynamics may be more complex than those captured by the HAR framework. Additionally, the LASSO(100) model demonstrates greater robustness to regime shifts than the HAR model, likely due to its ability to incorporate more relevant predictors. This adaptability suggests that regularized regression models may also capture the seasonality characteristic of realized volatility (C5).

Ding et al. (2021) further investigated the performance of the LASSO model relative to the HAR model in forecasting realized volatility. Using daily realized volatility data from eight international stock indices (UK FTSE, Japan N225, US SPX, Germany DAX, China SSEC, India NSEI, Brazil BSVP, and Mexico MXX) spanning 2006 to 2020, they evaluate performance using various forecast metrics, including *MSE*, *QLIKE*, *MCS*, and *VaR*. Their coefficient plots indicate that the first 22 lags are the most influential in terms of magnitude and statistical significance, lending some empirical support to the HAR model's lag structure. However, they also note that more distant lags still contain useful information, which contributes to the superior out-of-sample forecast accuracy of the LASSO \(AR(100)\) model over the HAR model.

Other studies have also examined the effectiveness of regularization techniques. Wei et al. (2020) analyze the use of ridge regression for forecasting the realized volatility of China's gold futures between March 18, 2011, and June 29, 2018. They find that Ridge regression


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## Kernel regressions

Another method related to the HAR model is kernel regression[9], a non-parametric technique. Kernel regression fits a simple linear model at each target point and weights observations based on their proximity to that point. This is done using a kernel (weighting) function $K_h(x_0, x_i)$ for $i=1,\ldots,N$, where *h* is a bandwidth parameter that defines the width of the local neighborhood. The kernel assigns weights to each observation $x_i$ based on its distance to the target point $x_0$. These memory-based methods require minimal training because the model essentially consists of the full dataset (Hastie et al., 2009, p. 191).

In its basic form, kernel regression solves a linear weighted least squares problem at each target point $x_0$. This involves minimizing the weighted sum of squared differences between observed values $y_i$ and those predicted by the local linear models. The optimization problem can be written as

$$\begin{aligned} min_{a_0,b_0} \sum ^N_{i=1} K_h(x_0, x_i)(y_i - a_0 - b_0 x_i)^2, \end{aligned}$$

(17)


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where $l_i(x_0)$ is a function of $K_h(x_0, x_i)$, $\hat{a}_0$ and $\hat{b}_0$, making the forecast, $\hat{y}_0$, linear in all $y_i$'s for $i=1,\ldots ,N$. Note that the kernel and the bandwidth parameter *h* are usually chosen using cross-validation (Hastie et al., 2009, p. 199). To apply kernel regressions to time series forecasting, the framework must be slightly adapted. In the context of realized volatility forecasting, LeBaron (2018) proposes a framework in which the kernel uses the same regressors as the HAR model. This design ensures that the kernel regression model can, at a minimum, capture the same characteristics of realized volatility as the HAR model, including autocorrelation (C1) and mean reversion (C4).

The kernel function assigns weights based on the distance between current and past average realized volatilities. Specifically, at time step *t*, it weighs $Y_t = [\tilde{\sigma }_{t-1}^{(D)}, \tilde{\sigma }_{t-1}^{(W)}, \tilde{\sigma }_{t-1}^{(M)} ]$ relative to past observations $Y_{t-i}$ for $i=1,\ldots ,t$. This approach effectively produces a weighted version of the HAR model, in which kernel regression searches for the past regime that most closely matches the present. Owing to its locality property, this framework is particularly suited to capturing regime shifts and seasonal patterns (C5).

Empirical results from LeBaron (2018), who modeled the realized volatility of the Dow Jones Industrial Index from 1933 to 1999, indicate that kernel regressions can capture nonlinear


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Şanlı et al. (2023) used kernel regression to forecast Bitcoin's realized volatility from September 2014 to March 2020, a period encompassing the COVID-19 pandemic. They recommended using exponentially weighted moving averages (EWMA) of past realized volatilities to enhance the kernel's locality features, as opposed to the simple averages employed by LeBaron (2018). Their model outperformed traditional alternatives based on the cumulative sum of squared prediction errors, particularly during regime shifts such as those triggered by the pandemic (Şanlı et al. 2023).

### Tree-based models

Another effective class of nonlinear models for time series forecasting is tree-based models. Unlike linear models, which require the specification of relationships between regressors and the target variable in advance, tree-based regressions adopt a non-parametric framework (Christensen et al. 2023). These models can naturally capture nonlinearities and interactions between explanatory variables. The core idea is to recursively partition the feature space into increasingly smaller subspaces using decision trees until a stopping criterion is met. Each terminal node of the tree provides a constant forecast for all observations that fall within that node. This approach is particularly effective at modeling complex data relationships and enables tree-based models to capture various characteristics of realized volatility (C1)–(C6). Owing to their nonlinear structure, these models are especially well-suited for capturing autocorrelation (C1) and mean reversion (C4). A basic


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1|t-5}\) represents the median of the 5-day average realized volatility. The numbers *split* [*obs*] represent the split point value and the number of observations that fall within this split

One limitation of tree-based regressions is that they pose a combinatorial optimization problem. There are no closed-form solutions for determining the optimal variable sorting and splitting order (Christensen et al. 2023). Because exhaustively evaluating all possible combinations is computationally intensive, most implementations rely on greedy algorithms that make locally optimal decisions to balance model accuracy and computational efficiency.

A popular and effective approach is the random forest (RF) model, which combines bootstrap aggregation (bagging)[10] with random feature selection for each split (Christensen et al. 2023). By constructing many computationally efficient regression trees and averaging their predictions, the RF model generates a final forecast that improves accuracy and reduces the risk of overfitting. Its key strength lies in its ability to combine numerous inaccurate but uncorrelated forecasts into a single, more accurate prediction.

Empirical results confirm the efficacy of RF models in forecasting realized volatility. For


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volatility, and Bouri et al. (2021), who examined Bitcoin volatility. Both studies confirmed the strong predictive performance of RFs, as indicated by standard metrics such as RMSE.

A recent advancement in tree-based models involves the application of boosting techniques. Unlike RFs, which generate trees independently using random subsets of features, boosted models construct trees sequentially. Each tree in the sequence is trained to correct the errors made by the previous ones. This iterative refinement improves model accuracy with each new tree. Boosting is particularly well suited for modeling complex and nonlinear datasets, offering improved forecasting accuracy based on RMSE and RMSE ratios relative to traditional RF models (Zhang 2022). In realized volatility forecasting, boosting also better captures long-term patterns such as seasonality (C5) than RFs. Two of the most effective tree-based boosting models used for this purpose are XGBoost, developed by Chen and Guestrin (2016), and LightGBM, developed by Ke et al. (2017).

Empirical evidence has shown the strong performance of tree-based boosting models in practice. Teller et al. (2022) used XGBoost to forecast the realized volatility of Dow Jones stocks from April 2007 to April 2019 and compared its performance to linear HAR models and LSTM networks. Their results show that XGBoost outperformed both models in one-step-ahead forecasts based on *MSE* and *QLIKE*, highlighting its ability to capture nonlinearities in short-term volatility. In another study, Wu and Wang (2021) found that


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## Deep learning models

This subsection examines models from the DL domain, which have recently emerged as promising tools for time series forecasting. These models are based on NNs and are characterized by their specific architectures. A key advantage of NNs is their foundation in the universal approximation theorem, which states that a NN can approximate any continuous function with arbitrary accuracy (Hornik et al. 1989). This capability allows NNs to model the full range of realized volatility characteristics (C1)–(C6) with considerable accuracy, giving them an advantage over traditional linear methods and many ML approaches that capture only subsets of these features.

In the time series literature, the two principal NN architectures are CNNs and LSTMs. We begin with CNNs and their applications in realized volatility forecasting, including a review of the CNN-based HARNet model. We then discuss LSTM models, including LSTM–CNN hybrids and transformers, and their effectiveness in volatility prediction.

### Convolutional neural networks

CNNs are a type of feed-forward NN that can effectively model complex time dependencies in time series data. They learn features by applying convolutional kernels with shared weights across the input, allowing them to automatically perform feature engineering. These kernels are moved across both time and feature dimensions, producing translation-


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A second advantage of CNNs is their capacity to identify localized patterns in asset and time dimensions. By applying convolutional kernels to a tensor containing multiple time series, CNNs can capture cross-asset effects over both short and long horizons. This allows them to model the co-movement characteristic (C6) of realized volatility (Borovykh et al. 2017).

A third benefit is the use of nonlinear activation functions in each layer, which enables CNNs to learn complex nonlinear relationships. This makes CNNs well suited to modeling autocorrelation (C1), asymmetry (C2), and leptokurtosis (C3). Unlike linear models, CNNs can capture nonlinear autocorrelations, offering greater flexibility in representing diverse temporal structures (Borovykh et al. 2017).

Lastly, CNNs are computationally efficient relative to many other DL models (Geras et al. 2016). Their design allows them to fit only the kernel parameters and apply the same weights across the input by sliding the kernel. This weight-sharing mechanism drastically reduces the number of model parameters compared to fully connected networks and improves generalization by mitigating overfitting (Goodfellow et al., 2016, p. 333).

Empirical studies have also demonstrated the success of CNNs in time series forecasting. Borovykh et al. (2018) found that CNNs are capable of capturing long-term dependencies and complex structures, and can process wide input ranges with fewer layers and


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The HARNet model, introduced by Reisenhofer et al. (2022), combines the traditional HAR model with CNNs. HARNet aims to integrate the volatility dynamics assumed in the HAR framework, particularly the persistence of daily realized volatility, with the high forecasting accuracy of CNNs. In its HARNet\(_{20}\) variant, the model uses the past 20 daily realized volatilities to predict the realized volatility for day \(t+21\). This design is inspired by the heterogeneous agent framework of the HAR model, which attempts to capture daily, weekly, and monthly volatility trends. A key innovation of HARNet is its use of dilated convolutional layers, which form a hierarchical structure that exponentially expands the model's receptive field without a corresponding increase in the number of parameters.

In this framework, the initial convolutional layer is crucial for detecting short-term volatility patterns by analyzing recent daily realized volatilities. As the data progresses through subsequent layers, the receptive field grows, enabling the model to learn long-term volatility dynamics. This progression from short- to long-term information is essential for accurately forecasting future volatility (Reisenhofer et al. 2022). Conceptually, the HARNet model can be interpreted as a nonlinear extension of the HAR model, where the contribution (weight) of each daily volatility observation is learned rather than prespecified.

Nonlinearities are incorporated into the HARNet model through the use of a rectified linear unit (ReLU)[12] activation function in each convolutional layer. These functions are essential


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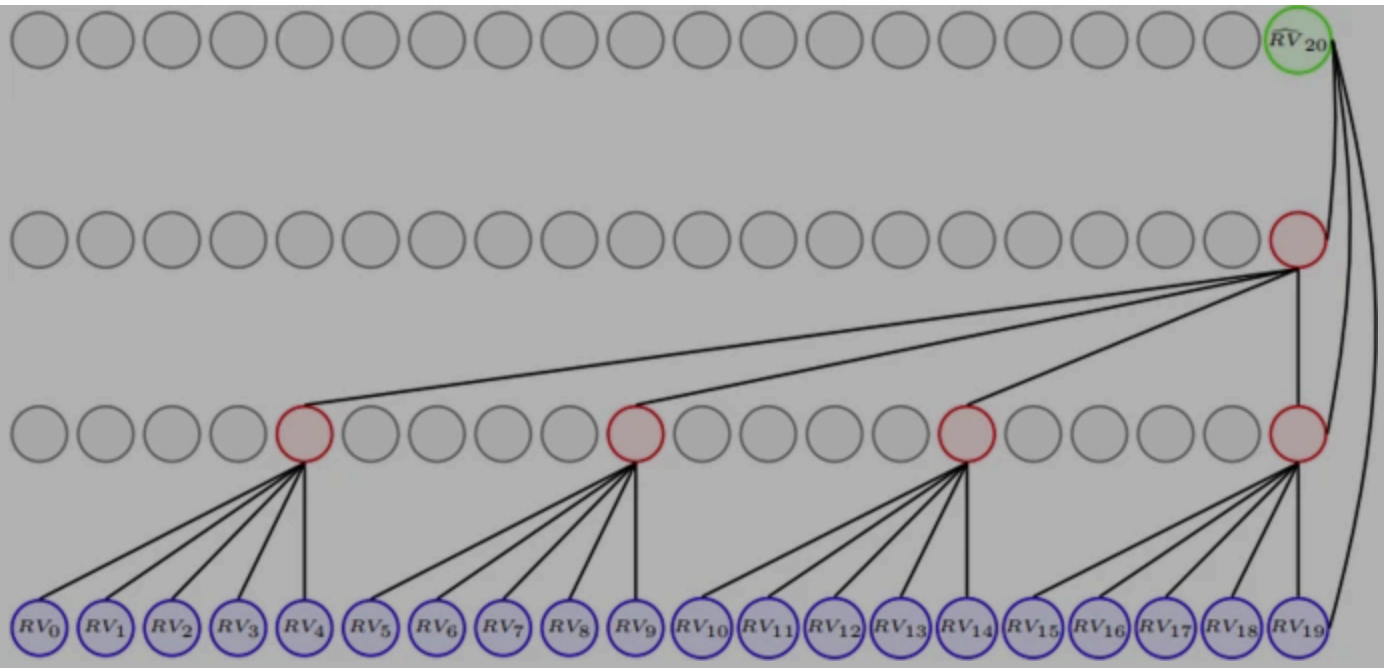

HARNet framework (Reisenhofer et al. 2022)

To mitigate common NN issues related to optimization and fitting, Reisenhofer et al. (2022) proposed a novel initialization technique. Their approach ensures that the initial forecasts of the HARNet model replicate those of the baseline HAR model by setting the initial weights of HARNet to the estimated coefficients from the HAR model. This is feasible because the model can be reformulated using convolution operations with average filters, revealing that it is a


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understanding of temporal volatility patterns and surpasses the HAR model in weighing past information for volatility forecasting (Reisenhofer et al. 2022).

### Long short-term memory networks

Another class of DL models that has proven effective in modeling time series is the LSTM network, introduced by Hochreiter and Schmidhuber (1997). LSTMs are a specialized form of RNN designed to address the limitations of traditional models in capturing long-term dependencies.[14]

The primary advantage of LSTMs lies in the architecture of their layers, which consist of memory cells and gating mechanisms. These input, forget, and output gates regulate the flow of information, allowing LSTMs to retain or discard information selectively across time steps. This functionality is critical in time series forecasting, where maintaining long-term context is often essential. For realized volatility forecasting, this allows LSTMs to effectively model seasonality (C5) and mean reversion (C4). Furthermore, the gated architecture mitigates vanishing and exploding gradient issues, thereby enabling the network to learn long-term dependencies over extended sequences without common optimization failures (Greff et al. 2016).

Like CNNs, LSTMs employ nonlinear activation functions, enabling them to model complex,


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These findings highlight LSTM's robustness in handling regime shifts. Furthermore, for multistep-ahead forecasts, they outperformed traditional models and matched the performance of other DL approaches in in-sample and out-of-sample contexts (Bucci 2020).

Similar findings were presented by Souto and Moradi (2023), who evaluated LSTM models for forecasting realized volatility during financial turbulence. Their study, which focused on the S&P 500 from November 2017 to November 2022, demonstrates that LSTMs effectively captured elevated volatility levels during market turmoil. This suggests that LSTMs accurately model the leptokurtosis of volatility (C3). When compared to econometric models such as GARCH and HAR, LSTMs yielded more accurate forecasts under volatile conditions, as measured by *RMSE* and *RMSE* ratios (Souto and Moradi 2023).

Further evidence comes from Vidal and Kristjanpoller (2020), who proposed a CNN-LSTM hybrid framework for forecasting gold price volatility. In their architecture, LSTMs captured short-term dependencies while CNNs modeled long-term patterns. Using daily realized volatility data from the London Bullion Market Association from April 1968 to October 2017, and an 80–20% train-test split, their model achieved a 37% reduction in *MSE* compared to the GARCH model and an 18% improvement over standalone LSTMs. The model also outperformed standalone CNNs, traditional NNs, and hybrid GARCH-NN models, highlighting the value of combining LSTM and CNN architectures. However, Ma et al.


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Transformers consist of an encoder and a decoder, each composed of multiple layers. The encoder processes the input time series and encodes it into a sequence of vectors. The decoder then uses these encoded representations to generate forecasts for future time steps. Both the encoder and decoder make use of multi-head self-attention layers. A single self-attention layer can be described as follows:

1. **Key, Query and Value Vectors:** For each data point $x_t$, the transformer layer calculates three vectors:

   - Key Vector: $K(x_t) = W^K x_t$
   - Query Vector: $Q(x_t) = W^Q x_t$
   - Value Vector: $V(x_t) = W^V x_t$

   where $W^K$, $W^Q$, and $W^V$ are learnable projection matrices. The key vector represents the encoding of the current time step. The query vector allows comparisons between the current time step and all prior time steps. The value vector represents the features of each time step, which are used in forecasting the next time step once the relevant ones are identified. These roles become clearer when steps 2 and 3 are taken into account.


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3. **Using Value Vectors:** Once attention scores are computed, the transformer layer uses the value vectors $V(x_j)$ to calculate the output. It aggregates these vectors, weighted by their corresponding attention scores, to produce the output of the self-attention layer:

$$\begin{aligned} Z_t = \sum _{j<t} A(x_t, x_j) V(x_j). \end{aligned}$$

(21)

Aggregating over $t=1, \ldots , T$, the resulting vectors can be stacked into a matrix $Z$.

One can stack *N* self-attention layers, each with independent random initializations of $[W_p^K, W_p^Q, W_p^V]$ for $p=1,\ldots , N$. Since each layer processes the same input $x_i$, this results in $Z_1,\ldots , Z_N$, which are then combined into a single output $Z^*$ using another learnable projection matrix $W^Z$. This transforms the self-attention layer into a multi-head self-attention layer, where $N$ denotes the number of *heads*. This design enables the model to simultaneously learn multiple representations of the data in the encoder and decoder layers, thereby enhancing the expressiveness of the model at each time step (Vaswani et al. 2017).


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making them less practical for small datasets. In such cases, CNNs often outperform transformers (Chen et al. 2022).

In the context of realized volatility forecasting, Frank (2023) evaluated temporal fusion transformers (TFTs)—a variant of the transformer architecture—using weekly and monthly realized volatility data from S&P 500 stocks between January 2000 and June 2022. The study found that TFTs, especially when incorporating pooling methods, significantly outperformed other DL and ML models, including LSTMs and Random Forests, based on RMSE and MAE. Moreover, Frank (2023) showed that incorporating sectoral pooling further improved forecast accuracy across all machine learning models.

These findings align with those of Ramos-Pérez et al. (2021), who proposed a multitransformer architecture that combines local and temporal transformers to capture both short- and long-term dependencies. Their study on the S&P 500 from January 2008 to December 2020 compared various neural network architectures. Hybrid models combining transformers and LSTMs outperformed all other approaches based on MAE and RMSE, particularly during high-volatility periods such as the 2020 COVID-19 pandemic.

Bringing together the insights from linear, ML, and DL models, we now turn to the conclusion, which summarizes our key findings and addresses the primary research


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To address these questions, we surveyed publicly available academic articles that referenced the term *forecasting realized volatility*, focusing on studies published between 2000 and H1 2024. Only studies that explicitly modeled realized volatility were included, excluding models where volatility was treated as a latent variable, such as those based on GARCH. Using this selection process, we reviewed the theoretical properties of each identified model. In total, we identified 32 unique models across 41 key articles. Each model incorporates specific empirical features of realized volatility and can be grouped into three broad categories:

1. *Linear models*: Traditional models such as ARIMA and HAR effectively capture basic volatility features like short-term autocorrelation (C1) and mean reversion (C4). Several extensions (e.g., A-HAR and asymmetric ARFIMA (C2); log-ARFIMA, log-HAR, and J-HAR (leptokurtosis (C3)); JA-HAR (asymmetry and leptokurtosis); T-HAR (seasonality (C5)); and VARFIMA and GHAR (co-movement (C6)), address additional characteristics. Nonetheless, these models often fall short in forecasting accuracy due to their limited ability to capture complex nonlinearities.

2. *ML models*: Machine learning methods such as regularized regressions, kernel regressions, and tree-based models offer improved accuracy by capturing nonlinear and complex temporal patterns. Regularized regression and kernel models are particularly strong in modeling short-term dependencies (C1), mean reversion (C4)


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Our findings show that DL models generally offer the most accurate forecasts of realized volatility by capturing complex nonlinear patterns and adapting to various market conditions. However, linear models remain important in settings with limited data or where interpretability is paramount. Models like GARCH and ARIMA continue to be widely used in practice due to their simplicity and low data requirements. In contrast, cutting-edge DL models such as Transformers require more memory and training time, as their parameters scale quadratically with input size.

The main theoretical contribution of this survey is that it offers the first direct comparison of traditional linear and ML models with state-of-the-art DL architectures, including CNN-LSTMs, HARNets, and Transformers. Practically, this comparison helps clarify the strengths and limitations of each modeling approach, supporting practitioners in selecting appropriate models based on forecasting goals, interpretability needs, and data availability. Accurate realized volatility forecasts are critical for risk modeling, monetary policy, and regulatory oversight. Our survey also highlights several promising directions for future research, including the development of hybrid models that integrate the strengths of different architectures, and improvements in the computational efficiency and scalability of DL approaches. Progress in these areas will further enhance volatility forecasting and expand its applications in financial risk management and derivative pricing.


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2. The implied volatility represents the market's expectation of future realized volatility and can be calculated from observed option market prices.

3. HAR and ARFIMA models.

4. Note that if $|d| < 0.5$, the ARFIMA model is both stationary and invertible (Kambouroudis et al. 2016). When $d \in (0, 0.5)$, the process displays long memory characteristics, while for $d \in (-0.5, 0)$, the process is characterized by antipersistence.

5. Aside from a constant, which is absorbed into the model's coefficients: $\alpha_1 \log(\tilde{\sigma}^2_t) = 2 \alpha_1 \log(\tilde{\sigma}^2_t) = \alpha^*_1 \log(\tilde{\sigma}_t)$.

6. Note that only trading days, when the exchange is operational, are considered and not calendar days.


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11. A receptive field is the area of input data that a single CNN filter covers.

12. Defined as \(f(x)=max(0,x)\). The ReLU adds nonlinearity to the layer by outputting zero for negative values and passing on positive values as they are.

13. The *QLIKE* error is computed by comparing the empirical quantiles to the quantiles forecast by the model. Lower *QLIKE* values imply better model performance, as they suggest that the predicted quantiles are closer to the actual quantiles.

14. Traditional RNNs have difficulties retaining long-term information over multiple time steps and suffer from vanishing gradients during training (Goodfellow et al., 2016, p. 398).

## Abbreviations

*A-HAR:* Asymmetric HAR


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*BSVP:* Bovespa index (i.e., a Brazilian stock market index)

*CAC:* Cotation assistée en continu (i.e., a French stock market index)

*CNN:* Convolutional neural network

*CSI:* China Securities Index (i.e., Chinese counterpart of the S&P 500)

*DAX:* Deutscher Aktienindex

*DJIA:* Dow Jones Industrial Average

*DL:* Deep learning

*EGARCH:* Exponential generalized autoregressive conditional heteroskedasticity

*ELBO:* Evidence lower bound

*EWMA:* Exponentially weighted moving average


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*IV:* Implied volatility

*IVOL:* Idiosyncratic volatility

*J-HAR:* HAR model with a jump component

*LASSO:* Least absolute shrinkage and selection operator

*LSTM:* Long short term memory

*MA:* Moving average

*MCS:* Model confidence set

*ML:* Machine learning

*MSE:* Mean-squared error

*MXX:* Mexico Index (i.e., a Mexican stock market index)


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*RA:* Autoregressive

*ReLU:* Rectified linear unit

*RF:* Random forest

*RNN:* Recurrent neural network

*RV:* Realized volatility

*SPX:* Standard and Poor's 500 index (i.e., an US stock market index)

*S&P 500:* Standard and Poor's 500 Index

*SSEC:* Shanghai Stock Exchange Composite (i.e., a Chinese stock index)

*SV:* Stochastic volatility

*SVM:* Support vector machine


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*VAR:* Vector autoregression

*VARFIMA:* Vector autoregressive fractionally integrated moving average

*VARMA:* Vector autoregressive moving average

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### Contributions

RML collected the articles and designed the survey framework, which was used to write the initial draft. NP reviewed and edited the manuscript, advised on changes, and supervised the research. All authors read and approved the final manuscript.

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## Competing interests

The authors declare that they have no competing interests.

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## Keywords

Realized volatility forecasting | Time series analysis | Linear models | Machine learning | Deep learning | Financial time series

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